NeoPixels

DO | PICO BLOCKS | OUTPUT | LIGHT

What does it do?

A NeoPixel is a full-colour LED with a tiny controller built into it.

One data wire carries colour instructions, and each pixel passes the remaining instructions onwards.

The 3.3 V-compatible JST breakout shown here can connect directly to a Pi or Pico.

Uses



Gaming PC lighting

DETECT Read lighting selection → **DECIDE** Which colour should each pixel be? → **DO** Light individual pixels



Disco shoelace meter

DETECT Read shoe movement → **DECIDE** Which light pattern should run? → **DO** Animate the pixels



Pixel RGB LED and controller

IN Data from the board

OUT Data to another pixel

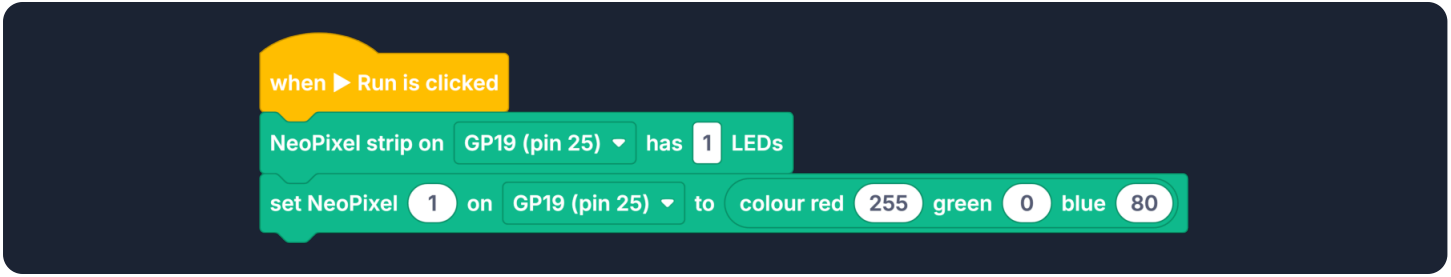
VDD and GND 3.3 V power

Raspberry Pi Foundation

Blocks

MicroLab

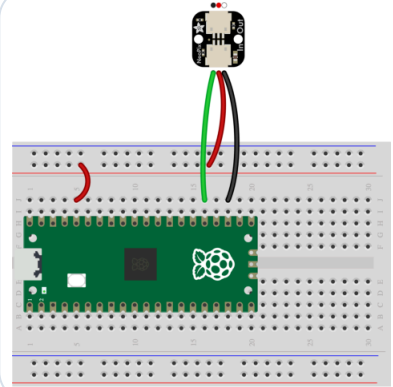
getmicrolab.com



Change the red, green and blue numbers (0 to 255) to mix a new colour.

Wiring

PICO



GP19 (pin 25) → IN, **3V3 OUT** (pin 36) → top + rail → VDD, **GND** (pin 23) → GND